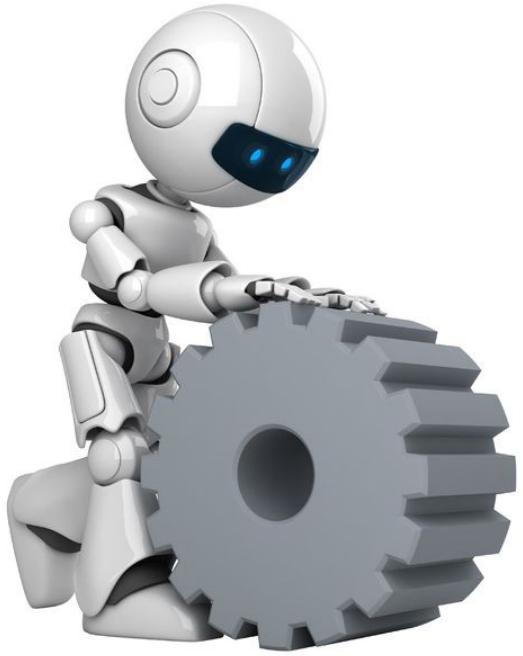
ICAT3110-3112

Intelligent Robotics



Lecture 2: Locomotion

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University of Vaasa

Location change on 10/2 and 12/2

Please check the schedule and location before lectures

Note that we always start **+15 minutes** past the lecture time to allow more participants to join!
E.g., lecture time at 14:00, the actual start at 14:15

Lecture No.	Date	Time	Location	Zoom Link
Lecture 1	26/1/2026	14:00-16:00	C303 Tervahovi	https://uwasa.zoom.us/j/63930259473?pwd=bhZzKyIc5hTG5TMDtEny6AEtCG1Hu2.1
Lecture 2	29/1/2026	14:00-16:00	C303 Tervahovi	https://uwasa.zoom.us/j/68780511364?pwd=dCbvCocQj2bWNhki8NS7eTSnFcVJza.1
Lecture 3	02/2/2026	10:00-12:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/64553376642?pwd=DTsPD5AhDL4DRZHu8WVEsw4avb2aQi.1
Lecture 4	04/2/2026	14:00-16:00	C203 Kurten Tervahovi	https://uwasa.zoom.us/j/65745013217?pwd=oosU6dvtnKDaasrHhu3o4aRjpypbIc.1
Lecture 5	10/2/2026	12:00-14:00	C203 Kurten Tervahovi	https://uwasa.zoom.us/j/67071083834?pwd=PtK77a3QSNVGPYwDYbx4IlYuzktMmB.1
Lecture 6	12/2/2026	10:00-12:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/64870927206?pwd=joc4JMVuy713ha7CKrkuCySOHW6qz3.1
Lecture 7	16/2/2026	10:00-12:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/62551103459?pwd=9E4O21JaoUuV8KqNIya2PbziYvzdw2.1
Lecture 8	18/2/2026	Group (A) 09:00-11:00 Group (B) 12:00-14:00	ABB Factory, Vaasa	**ABB Oy Factory Site Visit (No online meeting!)**
Lecture 9	24/2/2026	12:00-14:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/66579698602?pwd=sXvOvxfa5mXyclqRMpW6B9w3u6N8uU.1
Lecture 10	26/2/2026	10:00-12:00	C203 Kurten Tervahovi	https://uwasa.zoom.us/j/62280835656?pwd=uhsz5NDoAr5GloKt0njvRaVVwfLRf9.1
Lecture 11	02/3/2026	12:00-14:00	Lecture at SAMK, Pori	https://uwasa.zoom.us/j/61669450219?pwd=u75DuOhiuFsyJgVLFjRpmtaDyRnxTV.1
Lecture 12	05/3/2026	14:00-16:00	F118 Fabriikki	https://uwasa.zoom.us/j/66549239286?pwd=3Mhe9xqa9TRU0UuKGtRRmRBE9zk7DT.1

TextBook

- Materials and slides are **mainly** gathered through

'Introduction to Autonomous Mobile Robots'
by R. Siegwart & I. Nourbkhsh & D. Scaramuzza.

ISBN 0-262-19502-X

The MIT Press
London, England

You can find it in Tritonia Library as e-book

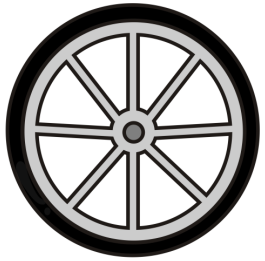
Locomotion

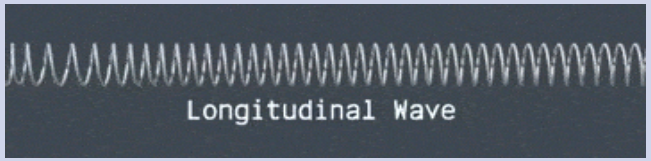
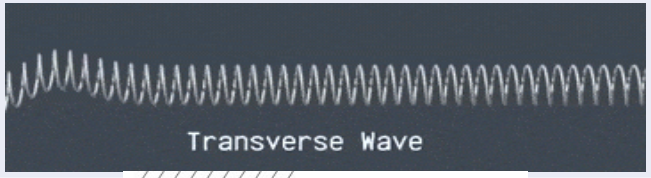
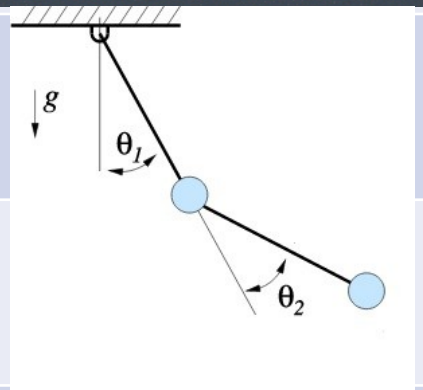
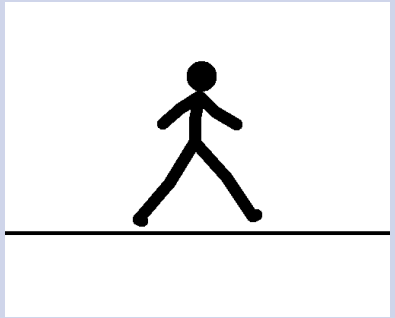
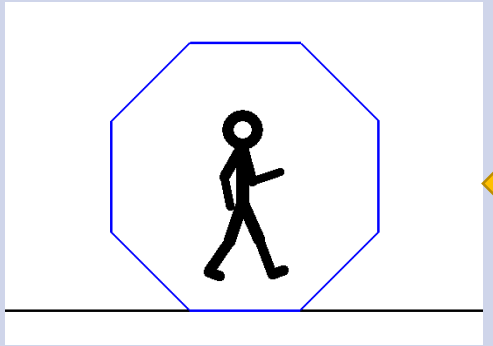
- A mobile robot needs locomotion mechanisms that enable it to move unbounded throughout its environment.
- Large variety of possible ways to move, and so the selection of a robot's approach to locomotion is an important aspect of mobile robot design.

- There are Robots that can walk, jump, run, slide, skate, swim, fly, and even roll.

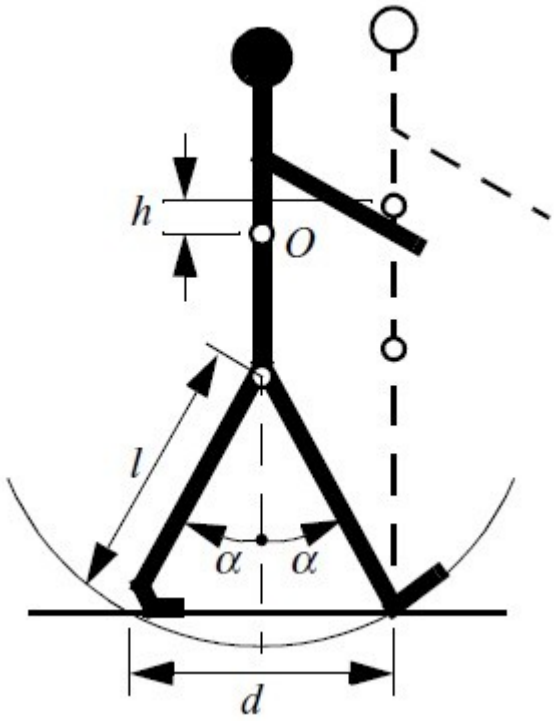
Most of these locomotion mechanisms have been inspired by biological counterparts i.e. Living Creatures like Mammals, Insects, Reptiles and Fish.

- One exception is the **Powered Wheel……** It's a Human invention.
- But it is similar to the **Human Bi-Pedal Walking system**….. **How ?**



Type of Motion	Resistance to Motion	Basic Kinematic of Motion
Crawling	Friction forces	
Sliding	Friction forces	
Running	Loss of Energy	
Jumping	Loss of Energy	
Walking 	Gravitational force Friction Forces	Biped walking 

Locomotion



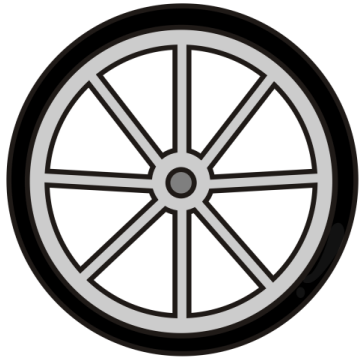
Bi-Pedal Walking System

Locomotion

- Mobile robots locomote either using wheeled mechanism or by number articulated legs, which is the simplest form of biological locomotion.

- **In General, Locomoted legs requires higher degrees of freedom, therefore gains more moving advantage over wheeled locomotion, yet complex mechanism.**

Vs



- Wheels are simpler locomotion form, but are only fine with **Flat grounds**.
- When it comes to **Flat ground**, legged locomotion ***loses***, Wheel ***wins***.

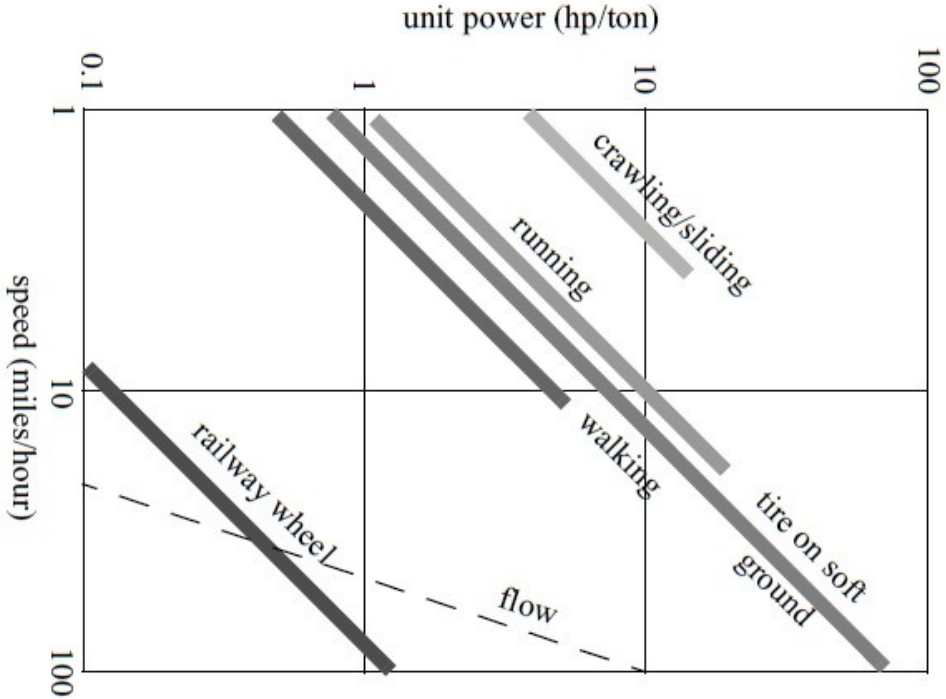
Locomotion

Railway is designed specifically for Wheeled locomotion,
Because on Hard-Flat steel surfaces, Rolling friction is minimal.

On the contrary, if wheels are rolled on Soft surfaces … Resistance is maximized.
In this case … **Legged Locomotion is better**, less resistant and more efficient.

From the graph:

- On soft grounds, wheeled locomotion suffers **higher** rolling friction.
- While legged locomotion suffers **less** friction.



Locomotion

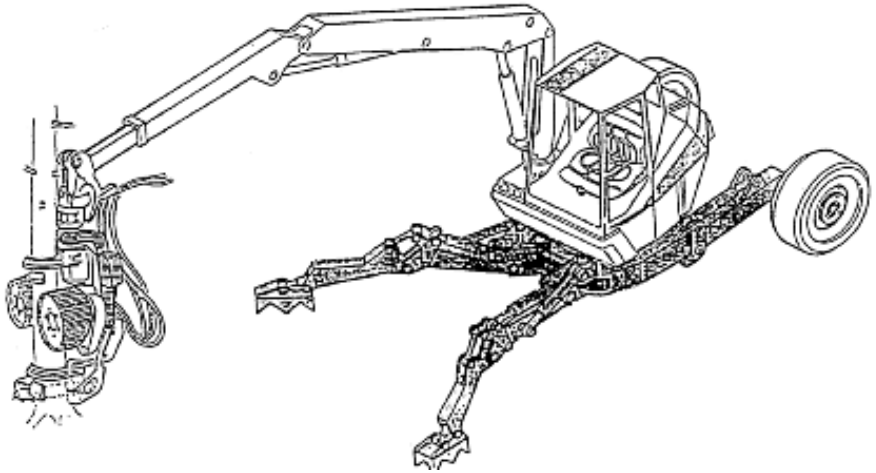
Conclusion

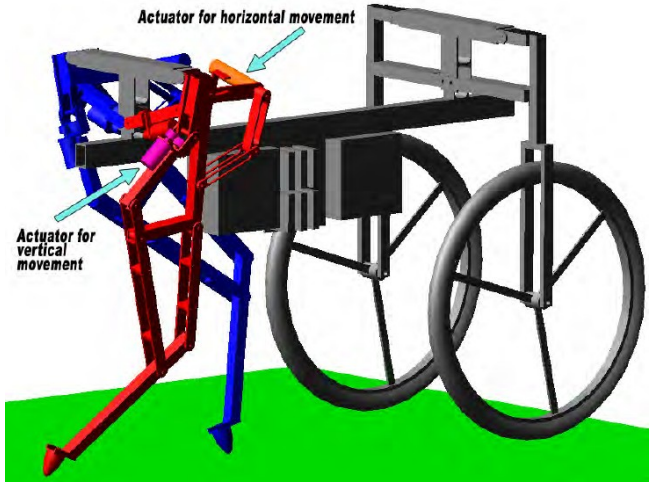
Nature favors Legged Locomotion !

- Wheeled Locomotion depends on Environment i.e. **Hardness and Flatness of surface**.
- Legged Locomotion depends on **Leg mass and Body mass** in which it should support.

Humans have to work on **flattening the surfaces** (Exert effort) to use Wheeled Locomotion.

Recently, more progress are done towards **combining both techniques together (Hybrid).**





Key issues of Locomotion

Locomotion is the complement of Manipulation.

Manipulation = Robot arm is fixed but moves objects in workspace.
Locomotion = Environment is fixed, Robot moves by imparting force to environment.

Both share core issues of

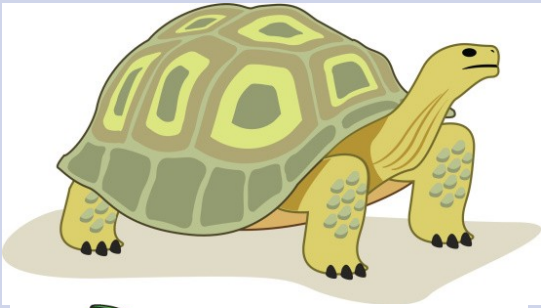
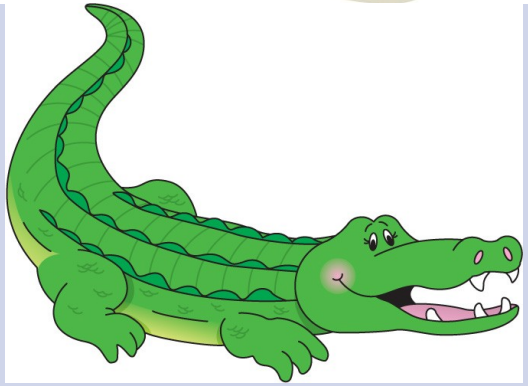
Stability	Characteristics of contact	Type of environment
- Number and geometry of contact points - Center of gravity - Static/dynamic stability - Inclination of terrain	- Contact point/path size and shape - Angle of contact - Friction	- Structure - Medium (e.g. water, air, soft or hard ground)

Legged Mobile Robots

- It characterizes by series of contact-points between the robot and ground.
- **Advantages**: Adaptability and Maneuverability in rough environment.
- Because quality of ground does not matter in such case as long as Robot maintain contact with ground through number of its points.
- **One more Advantage:** Can manipulate objects with great skill.
- **Disadvantages:**
  - Power and Mechanical complexity.
  - The leg, which include degrees of freedom, must be capable of sustaining part of the robot's total weight, and in many robots must be capable of lifting and lowering the robot.

Leg Configuration and Stability

- Legged Robots are biologically inspired, of course these systems are ***successful***.

Two or Four Legged Mammals	Four Legged Reptiles	Six Legged Insects
	 	

Leg Configuration and Stability

- Large Mammals and reptiles, have four legs whereas insects have six or more legs.

- In some mammals, the ability to walk on only two legs has been perfected. Especially in the case of humans, balance has progressed to the point that we can even jump with one leg,

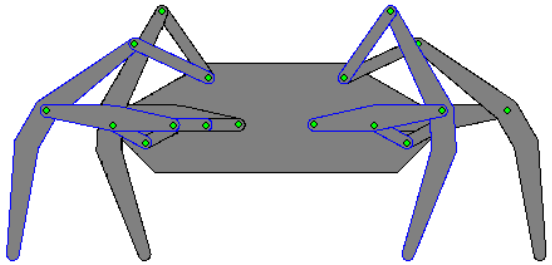
on the expense of control complexity.

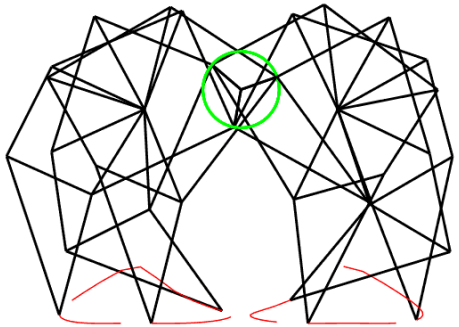
On the other hand

- A creature with three legs can S**tatic Stability** and adjust its center of gravity **e.g.(Tripod).**

For a Robot to posses Static Walking, it should have **at least Six legs.**
In such configuration, at least Three legs (**Tripod**) forming the walking gait is guaranteed

all the time !





Leg Configuration and Stability

- That's why Insects find it very easy to walk directly after being **Born**.
- While Mammals (Ex: Humans) should exert some effort learning how to stabilize their first steps in walking.

- New Born Mammals, with four legs, cannot achieve static walking, but are able to stand easily on four legs.
- They do the same as Humans, but in shorter time, they spend several more minutes/hours learning to walk without falling.

Leg Configuration and Stability

- Within each leg, there are a number of complexities, Once again, checking biological examples:

Caterpillar:

- Each leg is extended using hydraulic pressure by constricting the body cavity and forcing an increase in pressure.
- And each leg is retracted longitudinally by relaxing the hydraulic pressure, then activating a single tensile muscle that pulls the leg in toward the body.
- Each leg has only a single degree of freedom, which is oriented longitudinally along the leg.
- The caterpillar leg is therefore mechanically very simple, using a minimal number of extrinsic muscles to achieve complex overall locomotion.

But large number of Legs would only do it.

Leg Configuration and Stability

- At the other extreme, the human leg has more than 7 major degrees of freedom.
- Combined with further actuation at the toes.
- More than 15 muscle groups actuate 8 complex joints.
- For Mobile Robots, minimum of Two DOFs are required for lifting legs.
- For more complex maneuver, Third leg is added.
- Recent successes in the creation of bipedal walking robots have added a fourth degree of freedom at the ankle joint.
- The ankle enables more consistent ground contact by actuating the pose of the sole of the foot.

Leg Configuration and Stability

- In General, adding more DOF to Robot leg ***= Increases maneuverability***.
- Both augmenting the rang of terrain it can travel with variety of gaits.

- Additional actuators require energy and control, and they also add to leg mass, further increasing power and load requirements on existing actuators.

These additional joints and actuators are on the expense of Energy, Control and Mass.

Main Disadvantage of excessive DOF

- In multi-legged Robot
  The number of possible gaits ***depends*** on the number of legs.

- The ***gait*** is a sequence of **lift** and **release** events for each individual leg.

How many possible gaits for k legs ?

Leg Configuration and Stability

- For a Mobile Robot with ***k*** legs,

The total number of possible events ***N*** for a walking machine is given by the relation

$$N = (2k - 1)\,!$$

- Hence for a biped walker *k = 2,* there are 6 possible events

1	2	3	4	5	6
Lifting Right leg	Lifting Left leg	Release Right leg	Release Left leg	Lift both Legs	Release both Legs

- **Question:** For a Mobile Robot with 6 Legs

$$N = (12 - 1)! = 11!$$

39 916 800 possible event !

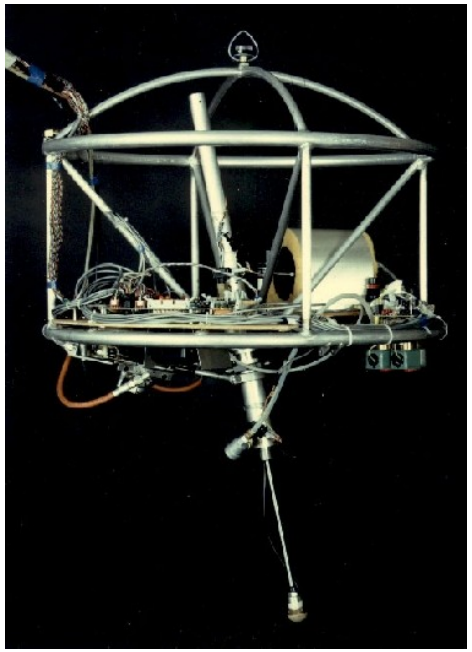
Legged Robots Examples

- As stated previously, for ensuring static and dynamic stability, minimum number of Robot legs should be **6** in order to provide **Tripod stability** for heavy duties.
- Of course there can be less than 6, but depends on the application, duty, weight and Robot design.
- There are no numerous major implementations of **Legged Robots** in industry **BUT** Researchers are focusing on developing this type of Robots for future use.
- Most of the existed Legged Robots are prototypes and under testing.
- Check Bristol Robotics Laboratory website for viewing these prototypes.

One-Legged Locomotion

- Of course it is the least number of legs in Legged-Robots.
- The Robot body should be designed in Light weight.
- No lifting duties should be assigned to such Robot.
- The contact area between the Robot and Ground is just 1 point.

……. What form can it takes to move ?

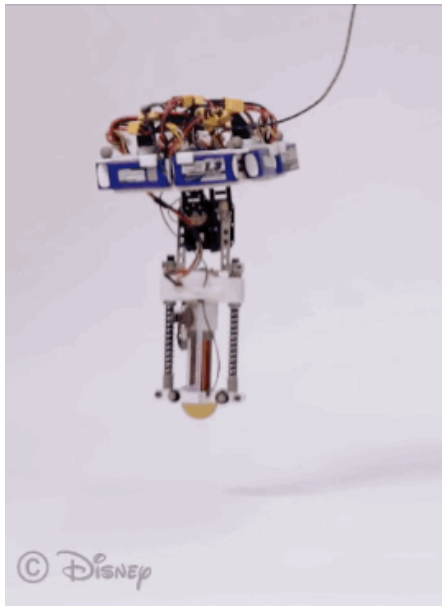


Raibert hopper

Benefits:

- No leg coordination is needed. Only needed in multi-leg Robots.
- Body mass is smaller and lighter weight
- Single contact point privileges the Robot the same as Wheels.
- For rough surfaces, a sequence of Single contact point can be the solution.

One-Legged Locomotion



Challenges :

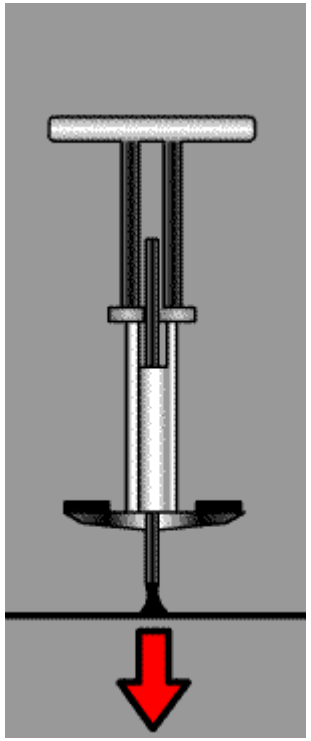
- Keeping balance!
- Static walking and stationery stability are ***nearly impossible !***
- Robot should adjust its center of gravity regularly.
- Or It should import corrective forces to correct its position.
- **To succeed = Dynamic stability should be ensured.**

Raibert Hopper:

- A well-known single-legged hopping robot.
- Designed by MIT (Massachusetts Institute of Technology) in **1983**.
- Built for experiments on active balance and dynamics in legged locomotion.
- Changeable Leg length, Carrying sensors and electronics.
- 2-axis hop actuators.
- Top recorded running speed **was 2.2 m/sec**

How it works ?

- Actuation is hydraulic. **Large off-board pump** should be connected all the time.
- High-power longitudinal extension of the leg during stance to hop back into the air.
- Designed to capture the kinetic energy of the robot as it lands, **using Springs**.
- This spring returns approximately **85% of the acquired energy,** the rest 15% energy needed are provided by hydraulic actuation.
- Can change speed by changing **angle of leg with respect to the body**.
- Can hop continuously for **20 minutes** with single battery set.



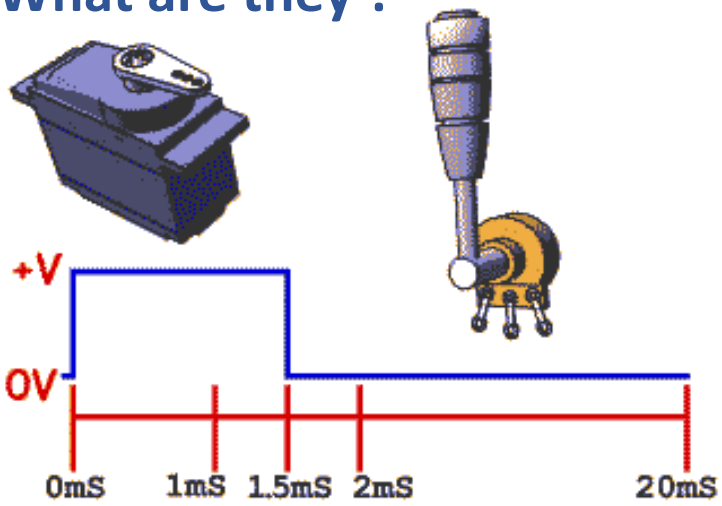
Two-Legged (Biped) Robots

- A variety of successful bipedal robots have been demonstrated over the past ten years. Two legged robots have been shown to run, jump, travel up and down stairways.

- HONDA and SONY have enabled highly capable bipedal robots.

- They designed small, powered joints that achieve **Power/Weight** performance in an efficient way than commercially available **Servomotors**. **What are they ?**

- These new “intelligent” servos provide not only **strong actuation** but also compliant actuation by means of torque sensing and closed-loop control.

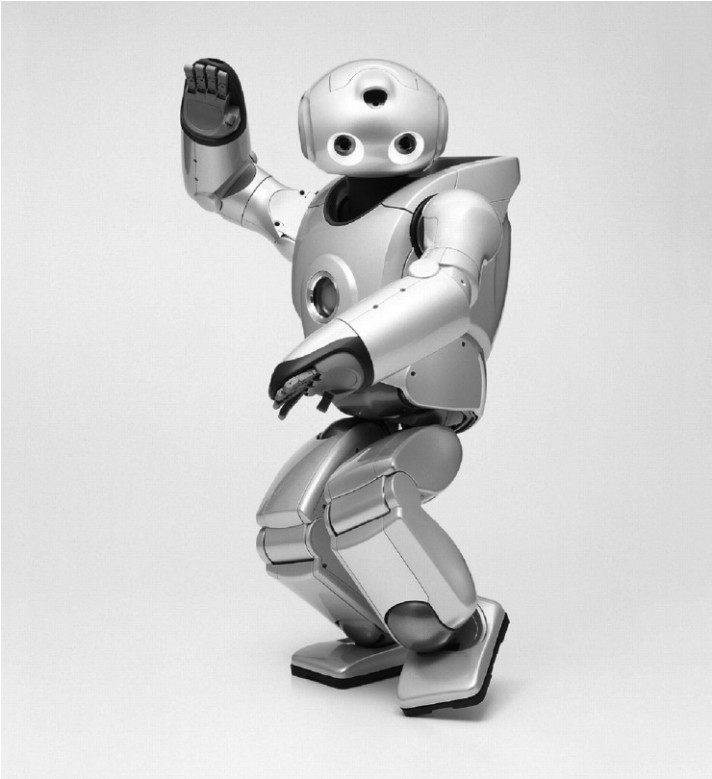


Two-Legged (Biped) Robots

Specifications:

Weight: 7 kg
Height: 58 cm
Neck DOF: 4
Body DOF: 2
Arm DOF: 2 x 5
Legs DOF: 2 x 6
Five-finger Hands

- Development started in 1997.
- Basic objective of motion entertainment and communication entertainment (i.e., dancing and singing).
- **Total 38 DOFs**



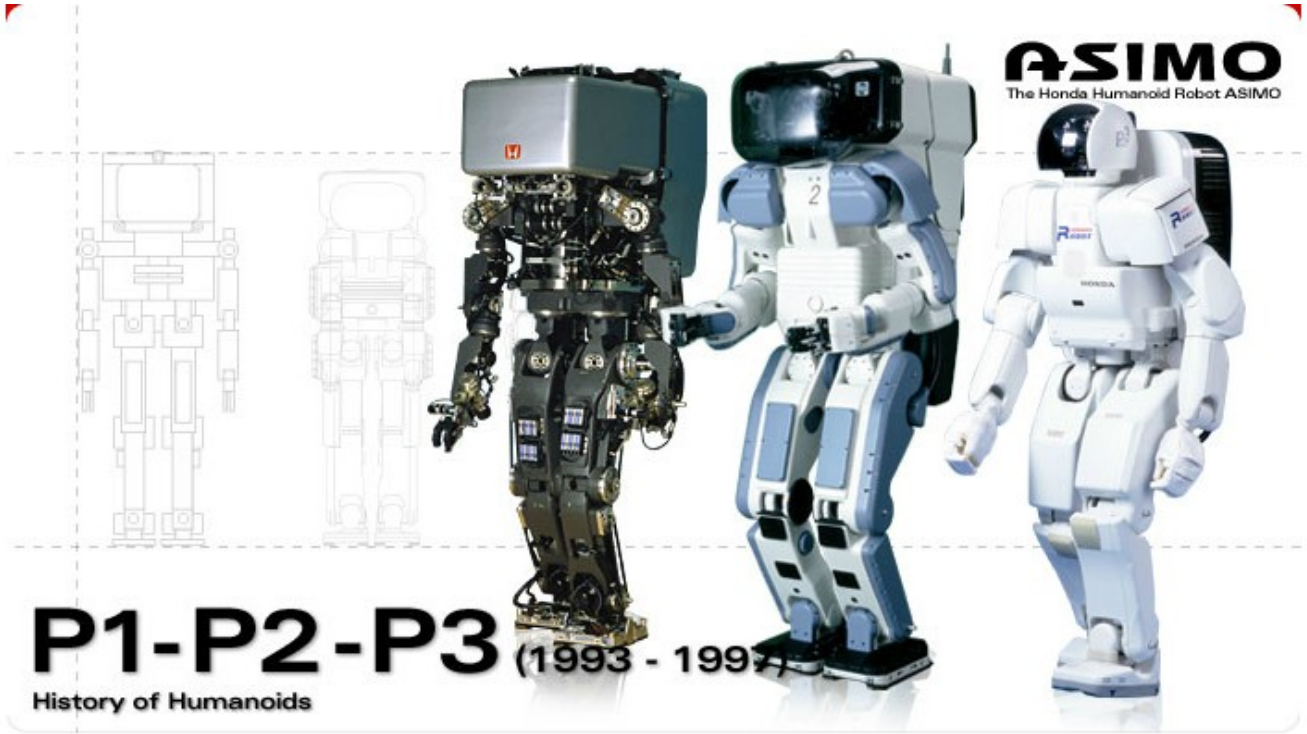
Sony SDR-4X

Two-Legged (Biped) Robots

- Equipped with **7 Microphones** for fine-localization of sound.
- Image-based person recognition.
- Limited ability to recognize human speech.
- SONY Engineers spent considerable effort on designing a motion prototyping application system to **script dances** in a **straightforward manner.**
- SDR-4X II the updated version of the robot was released in 2003.
- For more information and full specs visit SONY website : http://bit.ly/2mF9sJ6

Two-Legged (Biped) Robots

- HONDA Humanoid



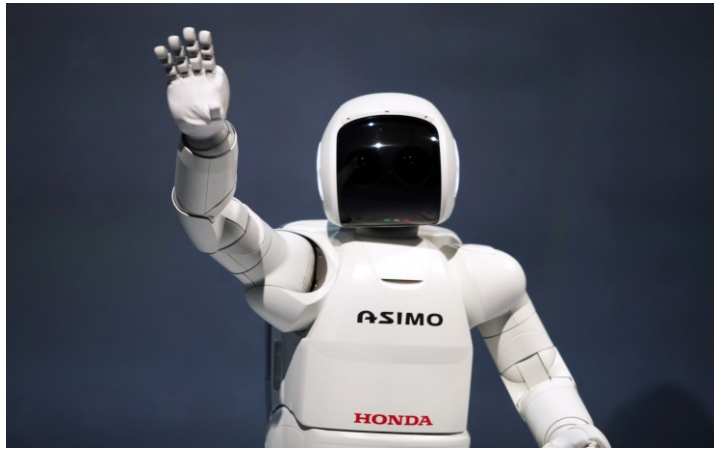
Specifications:

- Maximum speed: 2 km/h
- Autonomy: 15 min
- Weight: 210 kg
- Height: 1.82 m
- Leg DOF: 2 x 6
- Arm DOF: 2 x 7

- Apparently larger than SDR-4X
- First Robot to demonstrate human bipedal stair climbing and descending.
- **NOT** for entertainment purposes but as human aids throughout society.

Two-Legged (Biped) Robots

- HONDA refers, for instance, to the height of Asimo as the minimum height which enables it to nonetheless manage operation of the human world, **<u>for instance, control of light switches.</u>**

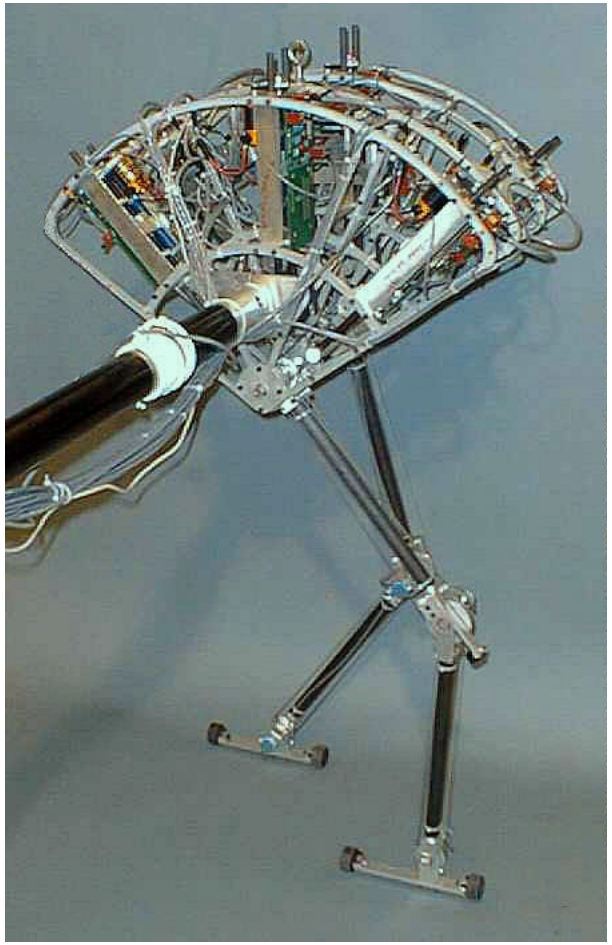


- An important feature of bipedal robots is their **anthropomorphic** shape.
- They can be built to have the same approximate dimensions as humans, and this makes them excellent vehicles for research in **human-robot** interaction.

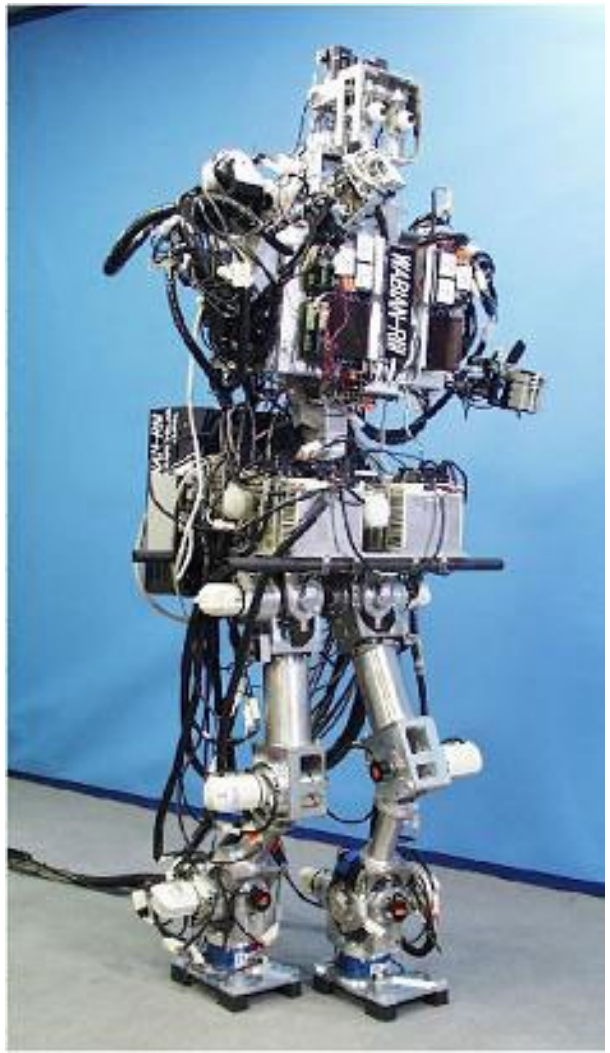
- Bipedal robots can only be statically stable **within some limits** Hence they should continuously perform **balance-correcting** by servoing even when standing still.
- For Two-legged Robots, each leg must have **sufficient capacity to support** the full weight of the robot.

Two-Legged (Biped) Robots

- **WABIAN** is a robot built at Waseda University-Japan for research.
- **WABIAN** is designed to emulate human motion, and is even designed to dance like a human.
- **Spring Flamingo** robot inserts springs in series with the leg actuators to achieve a more elastic gait.
- Limiting the joint angles using “kneecaps”, the Flamingo achieves surprisingly biomimetic motion.



Spring Flamingo



WABIAN-RIII

Video ….

Four-Legged (QuadRuped) Robots

- Standing still on four legs is **passively stable**, walking remains challenging!
- Because while walking, Center of Gravity should be shifted during the gait.
- **AIBO Robot** developed by SONY is one example.

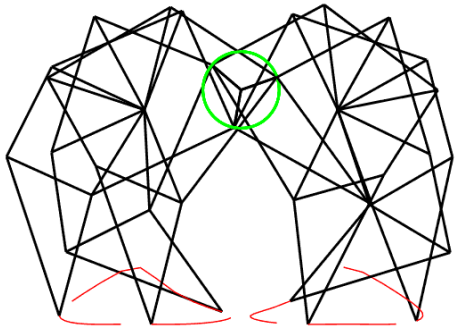
Video ….

- SONY produced a specific new OS especially for **AIBO** gait.
- The OS almost real-time, aided with new Servomotors.
- The high torque produced by these motors can support the robot during movement.
- SONY incorporated a **color vision system** that enables AIBO to chase a brightly colored ball !
- **AIBO** functions for full 1 hour, then it has to be recharged.
- 60,000 units sold in the first year.
- Humans can treat the Sony robot, **as a pet** and might develop an emotional relationship similar to that between man and dog.

AIBO

Six-Legged (Hexapod) Robots

- Six-legged configurations have been extremely **popular in mobile robotics** because of their **static stability** during **walking**, thus reducing the control complexity.
- In most cases, each leg has three **degrees of freedom**.



- **Genghis** a commercially available **hobby robot** that has six legs.
- Each legs has **2 DOF** actuated by Servomotors.
- Consists only of hip flexion and hip abduction, straightforward arrangement of servomotors and straight legs.

Hence less maneuverability in rough terrain.

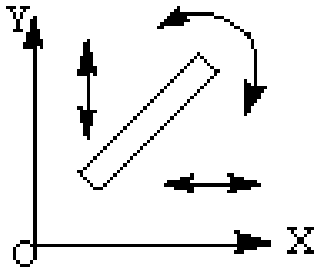
- Insects, are arguably the most **successful locomoting** creatures on earth.
- They are traversing all forms of terrain with six legs, even **upside down**.

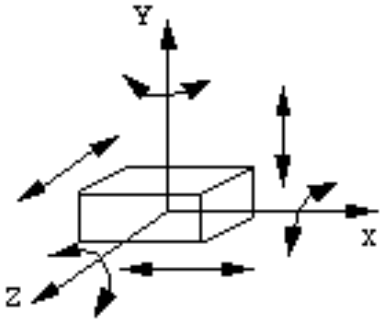
Genghis

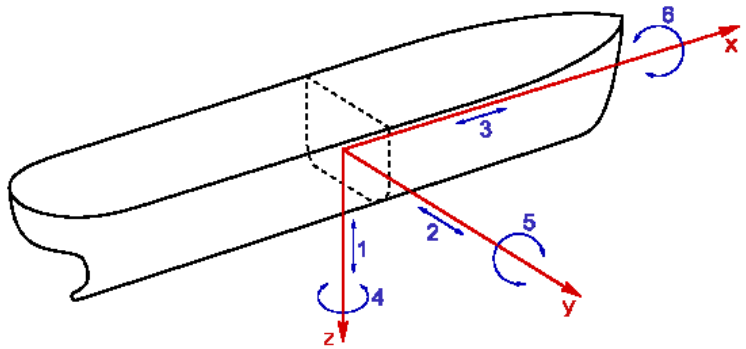
Six-Legged (Hexapod) Robots

Video ….

- *Recall* **DOF:** Degree of Freedom of a rigid body, the number of independent available movement it has.







- The gap between the capabilities of Insects and artificial robots is still **quite large**.

- This **is not** due to a lack of sufficient numbers of degrees of freedom on the robots.
- Rather, insects combine a small number of **active** degrees of freedom with **passive** structures**,** such as microscopic barbs and textured pads, that increase the gripping strength of each leg significantly.

Researches to attempt recreation of complete Mechanical functions of a **Cockroash !**

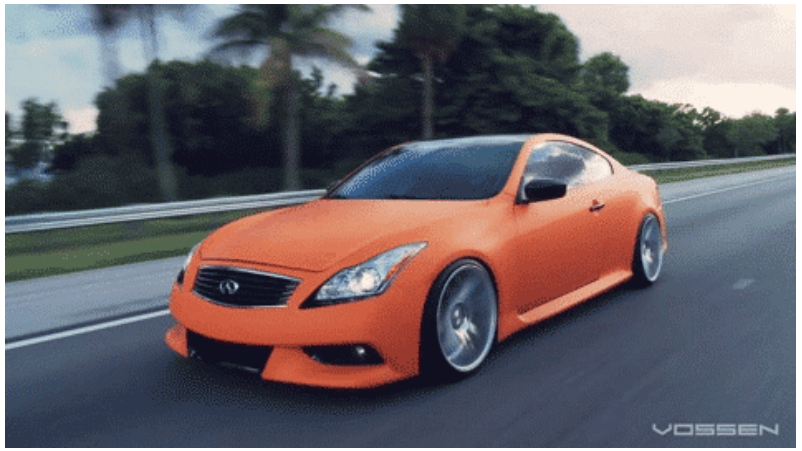
Those were Legged Robots

How about

Wheeled Mobile Robots?
?

Wheeled Mobile Robots

- By far it is the ***most popular locomotion*** mechanism in mobile robotics/vehicles designed by Human.



- Simple mechanism, very efficient in solid flat surface and keeps stability.
- Wheels are in **direct contact with ground** all the time so balancing is kept constant.
- **Three wheels** are sufficient to guarantee stable balance.
- **Two wheels** can also be stable BUT more effort and more control are needed.

Wheeled Mobile Robots

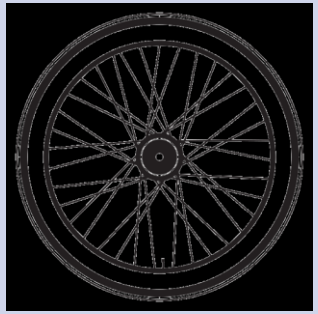
- Instead of worrying about balance, wheeled robot research tends to focus on the problems of **traction** and **stability**, **maneuverability**, and **control.**

- Can the Robot Wheels provide **sufficient traction and stability** for the robot to cover all of the **desired terrain?**
  Does the robot's wheeled configuration enable sufficient control **over the velocity of the robot**?
- There is a very large space of **possible wheel configurations** when considering possible techniques for mobile robot locomotion.

- **Firstly**, studying different **wheel types** with specific strengths and weaknesses.
- **Secondly**, examine complete **wheel configurations** that deliver particular forms of locomotion for a mobile robot.

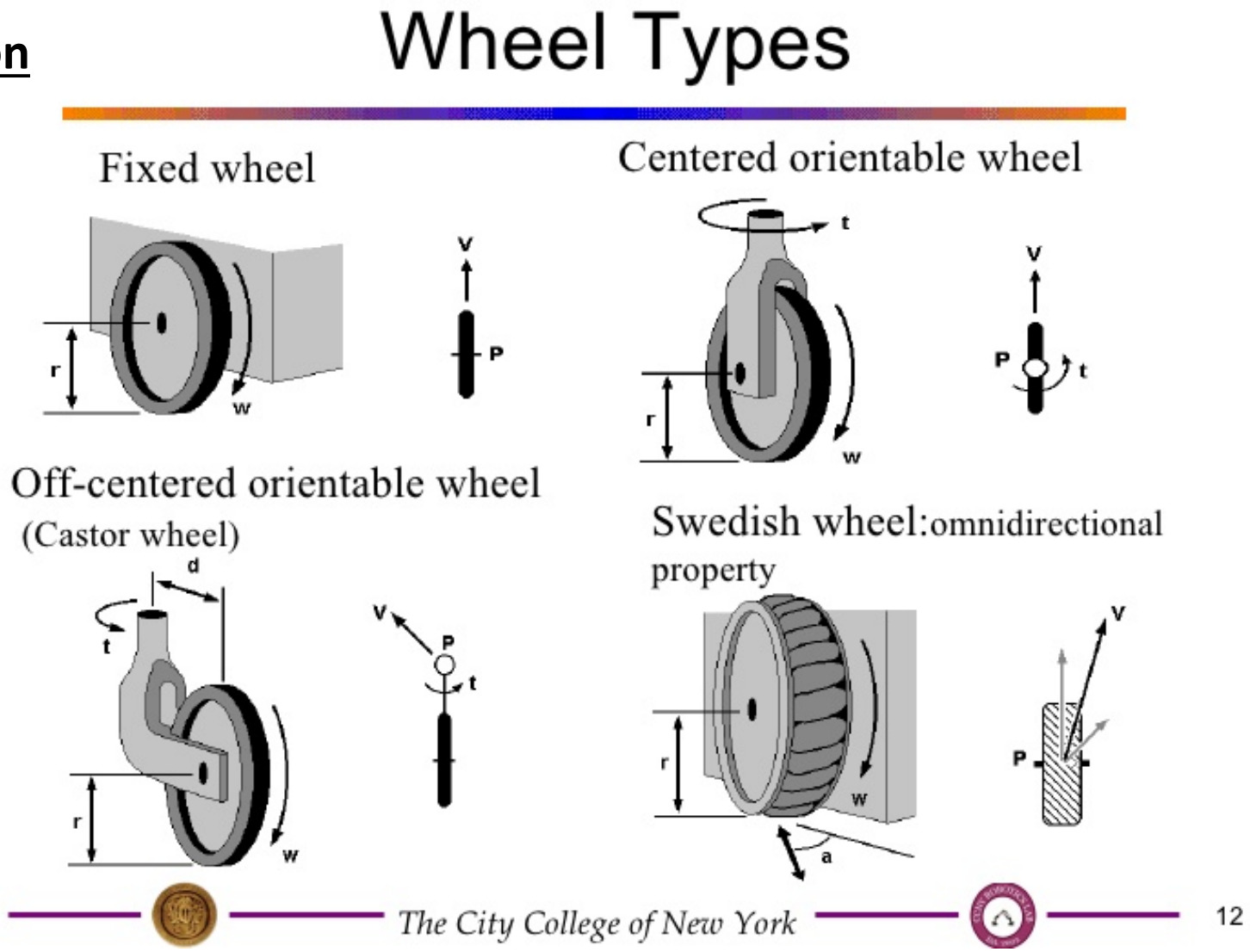
Wheeled Design

- There are **Four** major wheel classes.
- They differ widely in their kinematics, and therefore the **choice of wheel type** has a large effect on the **overall kinematics** of the mobile robot.

Standard Wheel	Castor Wheel	Swedish Wheel	Spherical Wheel
			
2 DOF Rotation around axis And contact point	2 DOF Rotation by steering joint And contact point	3 DOF Video …. Rotation around axis Rollers And contact point	**DOF:** Omni Directional **Ex:** Computer Mouse

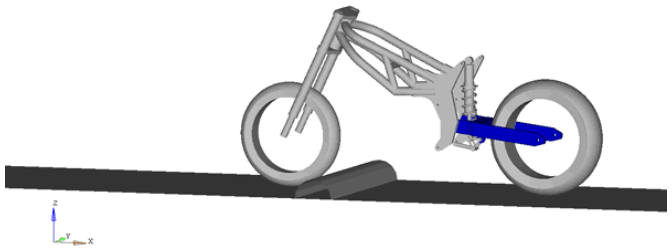
Wheeled Design

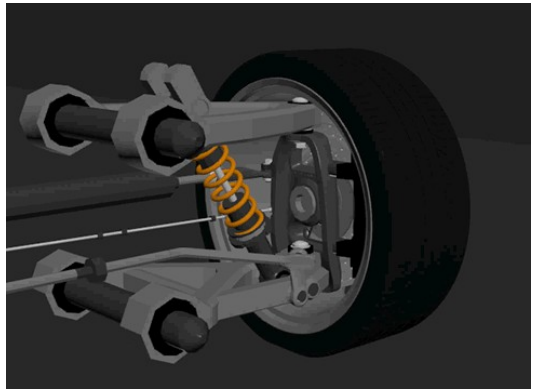
- **Another Classification**

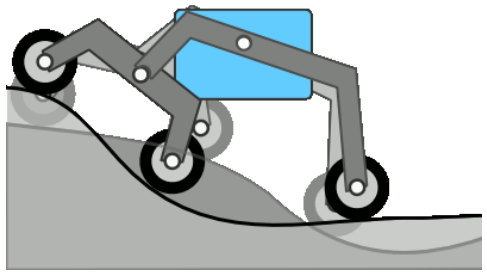


Wheeled Design

- For Robots which are designed for **all-terrain environments** and for robots with more than three wheels, a suspension system is normally required to **maintain wheel contact** with the ground.
- One of the simplest approaches to suspension is to design flexibility into the wheel itself. **Ex:** Manufacturers have applied a **deformable tire of soft rubber** to the wheel to create a primitive suspension.







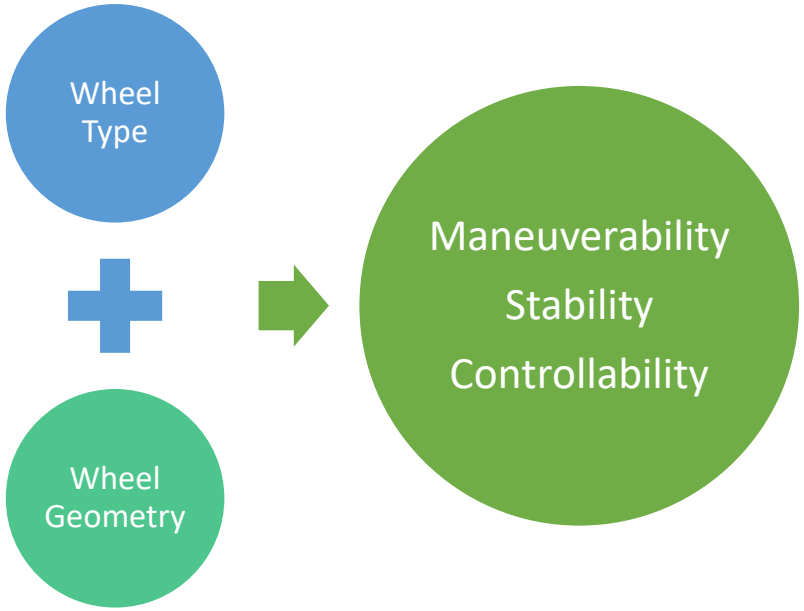
- Of course there should be more **sophisticated suspension** system in applications where the robot needs a more **dynamic suspension** for significantly **non-flat** terrain.

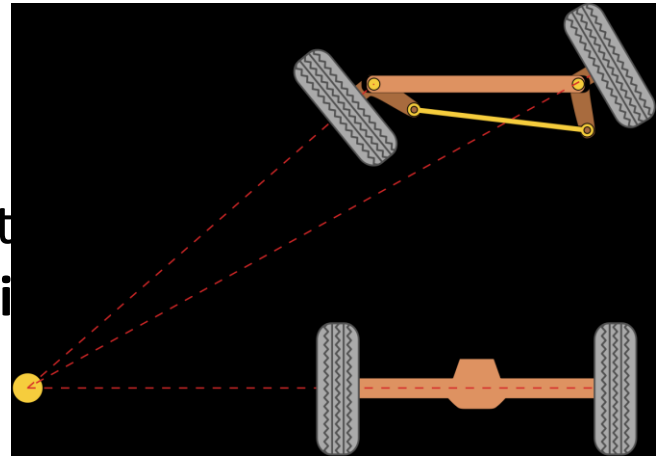
Wheeled Geometry

- Automobiles are designed for a highly standardized environment (**the road network**), mobile robots are designed for applications in a wide **variety** of situations.

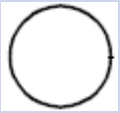
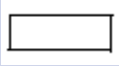
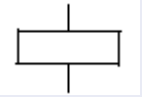
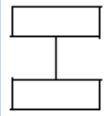
- Automobiles all share similar wheel configurations because there is **a region(s)** in the design space that maximizes maneuverability, controllability and stability for their standard environment: **the paved roadway.**

- The **Ackerman wheel** configuration of automobiles is not suitable for mobile Robots due to its **poor maneuverabili** unless the Robot is designed for road system as like cars.

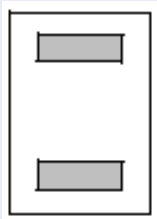
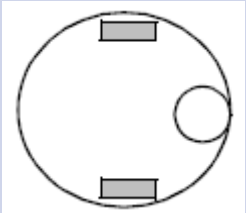
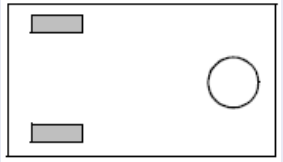




Wheeled Geometry

Icon	Description
	Unpowered omnidirectional wheel (spherical, castor, Swedish)
	Motorized Swedish wheel (Stanford wheel)
	Unpowered standard wheel
	Motorized standard wheel
	Motorized and steered castor wheel
	Steered standard wheel
	Standard connected wheels.

Wheeled Geometry

Number of Wheels	Arrangement	Description	Examples
2		One front **steering** wheel One rear **traction** wheel	Bicycle Motorcycle
		Two-wheel differential drive	Personal Cye
3		Two-wheel centered differential drive with a third point of contact	smartRob, EPFL
		Two independently driven wheels in the rear/front, 1 unpowered omnidirectional wheel in the front/rear	Pygmalion Many indoor robots

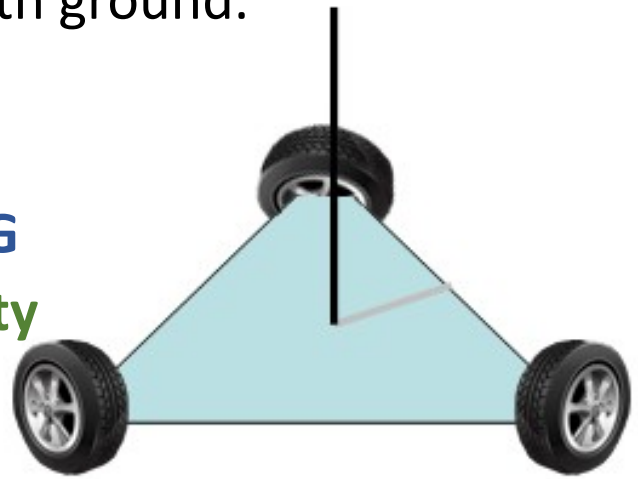
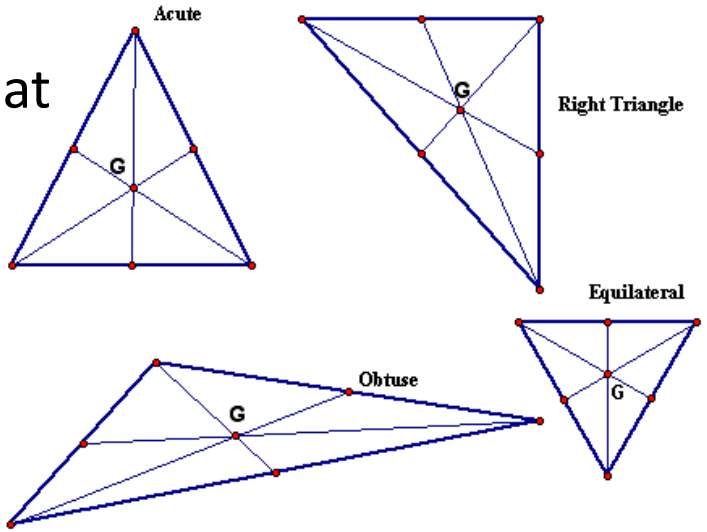
More ….

Wheeled Locomotion Stability

- Surprisingly, the minimum number of wheels required for **static stability is two**.
  <u>If the application is relatively small-sized and simple.</u>
- But in larger diameter wheels applications + High motor torque force.
  <u>Minimum Three wheels for static stability.</u> **And ?**
- Besides the number of wheels used, another important parameters is needed for static stability... **<u>Center of Gravity</u>**
- Center of Gravity should act within the contained triangle that is formed from contact points with ground.

Irregular shapes

Non-uniform weights

Causes shifts in COG

Adding more wheels boosts Stability